

Kinsey Reed Close, PhD

Postdoctoral Researcher · Microbial Ecology & Soil Health

West Virginia University · Morrissey Microbial Ecology, Biogeochemistry & Soil Health Lab · Morgantown, WV
kyreed@mail.wvu.edu | ORCID 0000-0001-5517-8664 | linkedin.com/in/kinseyreed | ResearchGate: Kinsey Reed Close

EDUCATION

- Ph.D., Plant & Soil Science** — West Virginia University DEC 2025
Emphasis in Applied & Environmental Microbiology · GPA 4.0. Dissertation: *Microbial and plant biodiversity in Appalachian agroecosystems and their implications for soil health.*
- B.S., Animal & Nutritional Sciences and B.S., Applied & Environmental Microbiology** — West Virginia University MAY 2019
Summa cum laude

POSITIONS HELD

- Postdoctoral Researcher, Morrissey Lab** — West Virginia University 2025–PRESENT
- GRASS Project Soil Health Specialist & Interim Program Coordinator** — WVU Extension 2024–PRESENT
Small Farms Center
- Ruby Distinguished Doctoral Fellow** — West Virginia University 2020–2024
- Farm Worker** — Rivendale Farm, Bulger, PA 2019
- Undergraduate Research Assistant, Freedman Lab** — West Virginia University 2017–2019

PUBLICATIONS

- Reed Close K**, Kubesch J, Matthias J, Basden T, Kotcon J, Felton E, Sperow M, Macias A, Guthrie K, Schartiger R, Winslett R, LeMaster D, Morrissey EM. Pasture plant community composition and structure in central Appalachia. **In preparation.**
- Reed Close K**, LeMaster D, Schartiger R, Guthrie K, Kane J, Kotcon J, Morrissey EM. 2026. Actinobacteria, mycorrhizae, and the biology of soil aggregate stability. *Soil Biology and Biochemistry* 213:e110048. doi.org/10.1016/j.soilbio.2025.110048
- Reed Close K**. 2025. Microbial and plant biodiversity in Appalachian agroecosystems and their implications for soil health. *Graduate Theses, Dissertations, and Problem Reports* 13096. doi.org/10.33915/etd.13096
- Kane J*, **Reed K***, Hofmockel K, Leshyk V, Morrissey E, Dijkstra P, Schwartz E, Sihi D, Hungate B. 2025. Advances in stable isotope profiling (SIP) techniques to track plant–microbial interactions. In: Norton J, Schimel J, Lindo Z, eds. *Measuring and Assessing the Biological Health of Soils*. Burleigh Dodds Science Publishing; Ch. 7, pp. 157–194. doi.org/10.19103/AS.2025.0159.07 (*Equal contribution)
- Reed K**, Dang C, Walkup J, Purcell A, Hungate B, Morrissey EM. 2025. Comparing field and lab quantitative stable isotope probing for nitrogen assimilation in soil microbes. *Applied and Environmental Microbiology* 91:e01849-24. doi.org/10.1128/aem.01849-24
- Reed K**, Morrissey EM. 2022. Bridging ecology and agronomy to foster diverse pastures and healthy soils. *Agronomy* 12(8):1893. doi.org/10.3390/agronomy12081893

GRANTS

Diversifying Appalachia's Pastures to Improve Soil Health — USDA NRCS Conservation Innovation Grant (Soil Health Demonstration Trial) \$1.0 million	2022–2026
GRazing Appalachia SuStainably (GRASS) — USDA Advancing Markets for Producers \$4.8 million	2024–2028

AWARDS & RECOGNITIONS

Nath Graduate Student Travel Award (\$1,500)	2025	Davis College Graduate Student Enhancement Grant (\$1,000)	2024
Oak Spring Garden Foundation Plant Conservation Biology Fellowship (\$10,000)	2025	Davis-Michael Scholarship (\$12,000)	2015–19
Ruby Distinguished Doctoral Fellowship (~\$150,000)	2020–23	WV STEM Scholarship (\$9,000)	2015–18
ASA-CSSA-SSSA Annual Meeting — 2nd Place PhD Oral Contest	2024	WV Promise Scholarship (\$19,000)	2015–19
3 Minute Thesis Finalist, WVU	2024	WV Farm Bureau Scholarship (\$2,000)	2016–18
Davis College Research Day — 2nd Place PhD Talk	2024	WVU President's List & Dean's List	2015–19
		Outstanding Senior in Plant & Soil Science; Outstanding Senior, WVU	2019
		Berkeley Springs High School Valedictorian	2015

INVITED TALKS & SEMINARS

- Reed K (Mar 2025). "Does plant diversity improve soil health & carbon storage in pastoral agroecosystems?" Oak Spring Garden Foundation, Upperville, VA.
- Reed K, Morrissey EM (Mar 2024). "Healthy Pastures, Healthy Soils, and Sustainable Grazing in Appalachia." Appalachian Grazing Conference, Morgantown, WV.
- Reed K (Feb 2024). "GRazing Appalachia SuStainably (GRASS) Project." West Virginia Small Farms Conference, Charleston, WV.
- Reed K et al. (Dec 2023). "Diversifying Appalachia's pastures to improve soil health." WV Stockman's College, Canaan Valley, WV.
- Reed K et al. (Nov 2023). "Diversifying Appalachia's pastures to improve soil health." WVU Extension ANR Unit Fall Meeting, Jackson's Mill, WV.
- Reed K et al. (Sep 2023). "Diversifying Appalachia's pastures to improve soil health." WV Conservation Agency, Deep Sea Acres Field Day, Pinch, WV.
- Reed K et al. (Aug 2022). "Diversifying Appalachia's pastures to improve soil health." Organic Farm Field Day, Morgantown, WV.
- Reed K, Morrissey EM (Mar 2022). "Plants, Microbes, and Soil Health in Appalachian Pastures." Appalachian Grazing Conference, Morgantown, WV.

CONFERENCE & ACADEMIC PRESENTATIONS

- **Reed Close K** et al. (Dec 2025). "Partnering with Appalachian farmers to diversify and manage pastures for increased soil carbon." [Oral] American Geophysical Union, New Orleans, LA.

- Aird A, Reed K, Morrissey EM (Apr 2025). "Exploring the Link Between Diversity and Aggregate Stability." [Poster] WVU Undergraduate Research Symposium, Morgantown, WV.
- Ozbolt R, Tripathi B, Reed K, Brzostek ER, Morrissey EM (Dec 2024). "Mycorrhiza–Saprotroph Interactions Shape Carbon Stabilization in Forest Soils." [Poster] AGU, Washington, D.C.
- Reed K et al. (Nov 2024). "Pasture plant community composition and structure in West Virginia, USA." [Oral] ASA-CSSA-SSSA Annual Meeting, San Antonio, TX.
- LeMaster D, Reed K, Morrissey EM (Nov 2024). "Actinobacteria and Soil Stability in West Virginia Pastures." [Oral] ASA-CSSA-SSSA Annual Meeting, San Antonio, TX.
- Bentley H, Reed K, Morrissey EM (Nov 2024). "Enzymatic Activity in West Virginia Forage Soils." [Poster] ASA-CSSA-SSSA Annual Meeting, San Antonio, TX.
- Reed K et al. (Apr 2024). "Pasture plant community composition and structure in West Virginia." [Oral] Davis College Research Day, Morgantown, WV.
- Reed K (Apr 2024). "Diversifying Pastures to Improve Livestock Production Sustainability." 3-Minute Thesis Competition, WVU.
- Reed K (Feb 2024). "GRazing Appalachia SuStainably (GRASS) Project." [Poster] Conservation and Agriculture Day at the Capital, Charleston, WV.
- LeMaster D, Reed K, Morrissey EM (Nov 2023). "How Grazing Intensity Impacts Soil Organic Matter and Aggregate Stability in West Virginia Pastures." [Poster] ASA-CSSA-SSSA Annual Meeting, St. Louis, MO.
- Reed K et al. (Mar 2023). "Diversifying Appalachia's pastures to improve soil health." [Poster] WVU Davis College Graduate Student Conference, Morgantown, WV.
- Morrissey EM, Dang C, Reed K, Brzostek ER, Walkup J, Pineiro J, Hungate B, Schwartz E (Nov 2021). "Connecting Microbial Biodiversity with Soil Carbon and Nitrogen Cycling Using Stable Isotopes." [Oral] ASA-CSSA-SSSA Annual Meeting, Salt Lake City, UT.
- Reed K (Oct 2021). "Diversifying West Virginia's Pastures to Improve Soil Health." [Departmental Seminar] Plant & Soil Science, WVU.
- Reed K, Morrissey EM (Aug 2021). "Advancing field-based techniques to identify microbial allies in nitrogen retention in agricultural soils." [Poster] Ecological Society of America Meeting.
- Reed K (2020). "Measures of Assessing Soil Nitrogen." [Departmental Seminar] Plant & Soil Science, WVU.

TEACHING

- **Courses:** Plant–Microbe Interactions (3 cr, co-taught; Biology 298C) & Biology 117 Honors Add-on (1 cr) — Fall 2026.
- **Guest Lectures:** Organic Farm Lecture Series, "Forage and grassland ecology" (2024); Environmental Microbiology — "Sulphur cycle" (2023), "Soil nitrogen cycling" (2022), "Bioengineering" (2022); Microbes & Global Change journal discussion (2021).
- **Tutoring:** Davis College of Agriculture & Natural Resources (2015–2019).

MENTORSHIP

- Undergraduate research mentees (WVU): Doreen Winter (2026–), Sapo Quintilla (2025–), Rebecca Ozbolt*† (2024–), Aidan Aird** (2023–25), Damon LeMaster*† (2022–24), Hannah Bentley** (2022–24), Abigail Judy‡ (2021), Akili Francist (2021).

* Resulted in 1+ research poster presentations † Entered/completed a graduate research program ‡ Working in a related industry or nonprofit position

LEADERSHIP & PROFESSIONAL SERVICE

- President & Founder, Plant & Soil Science Graduate Student Association (2021–2023); Board Member, Davis Graduate Student Organization (2022–2023); President & Founder, WVU Track Club (2016–2019).
- Co-reviewer, *Ecology*, Reviewer, Oak Spring Garden Foundation Plant Conservation Biology Fellowship Applications (2025) and Interdisciplinary Residencies & Fellowships Applications (2026).
- **Societies:** Ecological Society of America; American Geophysical Union; International Society for Microbial Ecology; ASA, CSSA, SSSA; Gamma Sigma Delta; Mortar Board; National Honor Society.

CERTIFICATIONS & TRAINING

- Certified Pharmacy Technician (WVBOP, 2020–Current); Hazardous Waste, Hazard Communication & Lab Safety (WVU EHS); CITI training (Research Security, Animal Handling, IACUC, Human Research, Agricultural Research & Teaching); Worker Protection Standard (WVU/EPA); Clery Act & Title IX; Purchasing Cardholder Ethics (State of WV).

REFERENCES

Dr. Ember Morrissey

Dissertation & Postdoctoral Advisor — Associate Professor of Biology, WVU

Dr. Zachary Freedman

Undergraduate Research Advisor — Assistant Professor of Soil Science, Univ. of Wisconsin–Madison

Kevin Jones

Farmer Research Cooperator — Owner-Operator, Deep Sea Acres Farm

Full reference contact details available upon request.

Dr. James Kotcon

PhD Committee & Collaborator — Emeritus Professor of Nematology, WVU

Dr. Edward Brzostek

PhD Committee & Collaborator — Associate Professor of Biology, WVU